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ACADEMIC DETAILS

Degree	Specialization	Institute	Year	CPI/%
B.Tech.	*Mechanical Engineering*	IIT Gandhinagar	2023-Present	8.02
Class XII	*Physics, Chemistry, Maths*	Prakash College	2021-2023	75.67 %
Class X		Ryan International School	2009-2021	96.28 %

RESEARCH EXPERIENCE

- **SketchWorld: Camera-Controlled Scene Generation from Sketches** [Aug '26 – Present]
  - Benchmarked **six controllable diffusion methods** on COCO and implemented **Universal Few-shot Control** on ControlNet 1.1, reaching **0.905 edge F1** from just **30 examples** of an unseen condition.
  - Engineered a **camera-controlled next-view forecasting pipeline** (**3.3 s** ahead) with SCEdit tuners on frozen **Stable Diffusion 1.5**, diagnosing it via ablations across **1,000 scenes**.
- **Multi-Parametric Retinal Diagnostics: Reproduction and Extension** | Project Link [Aug – Nov '26]
  - Reproduced a **physics-based OCT classification pipeline** from scratch on a **517-scan** benchmark, matching published results across all **four diagnostic classes**.
  - Exposed dataset confounds via **cross-scanner validation** on **3 OCT datasets** and permutation audits, benchmarking against a **fine-tuned CNN** for the accuracy-interpretability tradeoff.
- **Vision-Language Policy Learning & Teleoperation for Robotic Manipulation** | Project Link [Jan '25 – Jul '25]
  - Built a **sim-to-real pipeline** to deploy the **OpenVLA-7B vision-language-action model** on a **Franka FR3**, with custom MuJoCo environments for language-prompted pick-and-place and object stacking.
  - Developed **differential IK controllers** to collect synchronized image-language-action data for **LoRA fine-tuning** of OpenVLA; earlier built **vision-based teleoperation** of an Addverb Heal arm via RealSense, ArUco and **ROS**.

INTERNSHIPS

- **Agentic Multimodal RAG Knowledge Assistant – EY GDS Mumbai** | Project Link [May '26 – June '26]
  - Architected a self-correcting **multi-agent LangGraph** pipeline with a **reflection-driven feedback loop** and hybrid retrieval fusing **text-embedding-3-large** and **ColPali** embeddings across **4 document formats**.
  - Deployed it as **5 Docker Compose microservices** with a **FastAPI** backend, PostgreSQL and Pinecone, using **SHA-256** deduplication for idempotent ingestion.
- **Local RAG-Based Policy Assistant – Deloitte India** | Project Link [June – July '26]
  - Built a fully **on-premise RAG assistant** for HR policy queries using **Ollama** and **ChromaDB**, with OCR-backed **OKF** ingestion and a human-in-the-loop review step before indexing.

PROJECTS

- **Paged KV Cache & Continuous Batching Inference Engine** | Project Link [Mar – Apr '26]
  - Implemented a **vLLM-style paged KV cache** from scratch in **PyTorch** (block allocator, per-request block tables, **copy-on-write** prefix sharing) with a step-level **continuous batching** scheduler.
  - Benchmarked against static batching on Poisson-arrival traffic (**TTFT, TPOT, p50/p95/p99**) using a GPT-2-style model trained from scratch with a custom **byte-level BPE tokenizer**, served via async **FastAPI**.
- **Prompted Drywall Defect Segmentation using CLIP-Conditioned Models** | Project Link [April '26]
  - Built **CLIP-UNet with FiLM conditioning** from scratch for text-prompted crack/taping segmentation on **960 images**, reaching **0.503 mIoU / 0.645 Dice** and beating fine-tuned **CLIPSeg** by **5 mIoU points** at **22.5 ms**.

COURSE PROJECTS

- **Small Multimodal Experiments: CLIP Loss to VLM Adaptation** | Project Link [Oct '26]
  - Implemented **CLIP loss** from scratch; tuned only SmolVLM's connector, lifting digit accuracy **76%** $\rightarrow$ **98%**.
- **Small Models and Attention: Transformers from Scratch** | Project Link [Sep '26]
  - Built **causal self-attention** and a **ViT**; positional encoding gave an **82% drop** under patch shuffling.
- **SIFT-Guided Optical Flow for Lunar Descent Trajectory Reconstruction** | Project Link [Oct – Nov '26]
  - SIFT-seeded **Lucas-Kanade** tracking, reconstructing a **3D lander trajectory** from **143** image pairs.
- **Multi-Scale Template and Feature Matching for Scene Search** | Project Link [Sep – Oct '26]
  - From-scratch **NCC matcher**, **Harris/DoG** detectors and rotation-invariant voting, reaching **1.000 IoU**.
- **Invisible SVD-Based Digital Image Watermarking System** | Project Link [Sep '26]
  - Implemented **SVD from scratch** via Gram-matrix eigendecomposition to embed QR watermarks, sweeping **11 alpha strengths** and testing robustness to **Gaussian noise** and **JPEG**.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++
- **ML & CV:** PyTorch, TensorFlow, Hugging Face, Stable Diffusion, ControlNet, OpenCV, scikit-learn, NumPy, Pandas
- **LLM, Systems & Robotics:** LangGraph, FastAPI, PostgreSQL, ChromaDB, ollama, ROS, MuJoCo ,Linux, Git, GitHub, Docker

POSITIONS OF RESPONSIBILITY & OUTREACH

- **Senior Designer, EII, IITGN** – Led a team of **7 designers** creating posters and merchandise for entrepreneurial events.
- **Hostel Activity Team, IITGN** – Organized community activities for **200+ residents** in a **3-member team**.
- **Teaching Volunteer, Nyasa, IITGN** – Taught **25+ students** in a student-led outreach initiative.